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EcoFleet AI: Intelligent Electric Vehicle Fleet Monitoring and Route Optimization Platform

SCENARIO-BASED REQUIREMENT ANALYSIS

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ABSTRACT

The global shift toward electric vehicles (EVs) has created a pressing need for intelligent fleet management systems that go beyond simple GPS tracking. Conventional fleet software tracks vehicle location but offers no insight into battery degradation, charging efficiency, or energy-optimal routing leaving fleet operators reactive rather than proactive. EcoFleet AI is proposed as an AI-powered platform that continuously monitors EV battery health, energy consumption, and vehicle status through IoT sensors, while using machine learning to predict battery degradation and maintenance needs before they cause downtime. The platform optimizes charging schedules around electricity demand, tariff windows, and charging-station availability, and recommends energy-efficient routes using live GPS and traffic data. Fleet managers are given real-time dashboards and analytical reports to support data-driven decisions, drivers receive optimized navigation and timely alerts, and technicians receive predictive maintenance tickets before failures occur.

The system is designed as a modular, cloud-hosted platform, allowing it to scale from small pilot fleets of a few dozen vehicles to large-scale deployments spanning thousands of vehicles across multiple depots and regions. Its layered architecture separates data collection (IoT telemetry), intelligence (machine learning prediction and optimization), and presentation (dashboards, mobile apps, and alerts), so that each layer can evolve independently as sensor technology, routing algorithms, and charging infrastructure mature. By unifying predictive maintenance, smart charging, and route optimization into a single cloud-based platform, EcoFleet AI aims to extend battery life, reduce energy and operational costs, increase vehicle availability, and support organization-wide sustainability and carbon-reduction goals for logistics, public transportation, and last-mile delivery fleets. In doing so, it positions itself not merely as a monitoring tool but as a decision-support system that helps fleet operators move from cost-reactive management to proactive, data-driven, and environmentally responsible fleet operations.

Keywords: Electric Vehicle Fleet Management, Predictive Maintenance, Route Optimization

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1. INTRODUCTION

1.1 Purpose

This document specifies the software requirements for EcoFleet AI, an AI-enabled platform for monitoring electric vehicle (EV) fleets and optimizing their charging and routing operations. It defines what the system must do (functional requirements) and how well it must perform (non-functional requirements), and serves as the baseline reference for design, development, testing, and evaluation.

1.2 Business Domain

The system operates in the smart transportation and green logistics domain, at the intersection of IoT (vehicle telemetry), AI/ML (predictive analytics), and cloud-based fleet operations software. It targets organizations operating EV fleets — logistics providers, public transit authorities, and last-mile delivery services — that need to convert raw vehicle data into actionable, cost-saving decisions.

1.3 Intended Audience

This SRS is intended for the software development team, project evaluators, fleet management stakeholders, quality assurance testers, and future maintainers of the EcoFleet AI platform.

1.4 Definitions and Abbreviations

Term	Meaning
EV	Electric Vehicle
SoC	State of Charge — current battery percentage
SoH	State of Health — indicator of battery degradation over time
IoT	Internet of Things
ML	Machine Learning
API	Application Programming Interface

Term	Meaning
SRS	Software Requirement Specification
GPS	Global Positioning System
MQTT	Message Queuing Telemetry Transport (lightweight IoT protocol)
KPI	Key Performance Indicator
SLA	Service Level Agreement

2. Problem Description

Electric vehicle fleet operators currently rely on conventional fleet management systems that primarily perform GPS-based vehicle tracking. These systems lack the intelligence to predict battery degradation, optimize charging schedules around electricity demand and station availability, or recommend energy-efficient routes. As a result, fleets suffer from unplanned downtime due to unexpected battery failures, unnecessary charging stops that delay deliveries, higher-than-necessary energy costs, and poor real-time visibility into vehicle and charging-infrastructure status. Fleet managers are forced to make decisions reactively, often after a problem has already impacted delivery schedules or vehicle availability. EcoFleet AI addresses this gap by combining IoT sensor data, GPS/traffic feeds, and machine learning models into a single platform that gives fleet managers predictive, actionable insight instead of raw tracking data alone — shifting fleet operations from reactive maintenance to proactive, data-driven management.

3. Scope of the System

3.1 In-Scope Features

- Real-time monitoring of battery health, State of Charge (SoC), State of Health (SoH), and vehicle status via IoT sensors
- ML-based prediction of battery degradation and maintenance scheduling
- Charging schedule optimization based on battery status, electricity tariff/demand, and station availability
- Route recommendation engine using live GPS and traffic data, optimized for energy efficiency
- Fleet manager dashboard with real-time visualizations and map views
- Analytical and utilization reporting with export capability
- User and role management for fleet managers, drivers, technicians, and administrators
- Notification/alert system for critical battery, maintenance, and charging events
- Charging session logging and historical audit trail
- Vehicle profile and service-history management

3.2 Out-of-Scope Features

- Autonomous vehicle control or self-driving integration
- Direct manufacturing/hardware production of EVs or charging stations
- Financial/billing settlement between the fleet operator and public charging networks
- Passenger-facing mobile applications for public transit riders
- Cross-fleet vehicle-sharing marketplace features
- Insurance claims processing integration

4. STAKEHOLDERS AND BUSINESS CONTEXT

Stakeholder	Role / Interest in the System
Fleet Manager	Monitors fleet health, approves charging/route plans, reviews reports, manages vehicle profiles
Driver	Receives optimized route guidance, vehicle alerts, and charging instructions
Maintenance Technician	Receives predictive maintenance alerts, logs completed service actions
System Administrator	Manages user accounts, roles, permissions, and system configuration
Charging Station Operators (external)	Provide real-time station availability data via API integration
Business/Organization Owner	Reduces operational cost, meets sustainability and emissions targets
Regulatory Body (indirect)	Concerned with data privacy and environmental compliance

Business Context: Fleet operators are under commercial pressure to reduce cost-per-kilometer and regulatory pressure to cut emissions. EcoFleet AI positions itself as the analytical layer that connects vehicle-level telemetry to fleet-level business decisions.

5. FUNCTIONAL REQUIREMENTS

ID	Requirement
FR-1	The system shall collect real-time battery health data (SoC, SoH, temperature, voltage) from IoT sensors installed on each vehicle.
FR-2	The system shall continuously monitor energy consumption and vehicle operational status (idle, in-transit, charging, faulted).
FR-3	The system shall use a machine learning model to predict battery degradation trends and estimate remaining useful battery life.
FR-4	The system shall generate predictive maintenance alerts before a vehicle component is likely to fail.
FR-5	The system shall optimize charging schedules based on current battery status, electricity demand/tariff windows, and charging station availability.
FR-6	The system shall recommend energy-efficient routes using GPS location, live traffic conditions, and vehicle performance/consumption data.
FR-7	The system shall provide a real-time dashboard displaying vehicle location, battery status, and fleet-wide KPIs to fleet managers.
FR-8	The system shall generate analytical reports on fleet utilization, energy efficiency, and operational performance over selectable time periods.
FR-9	The system shall send real-time alerts/notifications to relevant users (drivers, managers, technicians) for critical events such as low battery, predicted failure, or charging station unavailability.
FR-10	The system shall allow fleet managers to view and manage vehicle profiles (registration, assigned driver, capacity, service history).

ID	Requirement
FR-11	The system shall allow administrators to manage user accounts and role-based access permissions.
FR-12	The system shall log all charging sessions with timestamps, energy consumed, and cost for auditing purposes.
FR-13	The system shall allow drivers to view their assigned route and receive turn-by-turn energy-optimized navigation guidance.
FR-14	The system shall integrate with external charging station systems to retrieve real-time station availability and slot booking status.
FR-15	The system shall support data export (CSV/PDF) of reports for offline review and compliance documentation.
FR-16	The system shall allow fleet managers to override or manually adjust AI-generated charging and route recommendations.
FR-17	The system shall maintain an audit log of all maintenance tickets, including technician actions and resolution timestamps.
FR-18	The system shall support geofencing so that alerts are triggered when a vehicle deviates from its assigned route or operational zone.
FR-19	The system shall allow administrators to configure alert thresholds (e.g., minimum SoC before a low-battery alert is triggered).
FR-20	The system shall provide a historical trend view of battery SoH degradation per vehicle for long-term fleet planning.

6. NON-FUNCTIONAL REQUIREMENTS

Category	Requirement
Performance	The system shall process and reflect IoT sensor updates on the dashboard within 5 seconds of data transmission. Route recommendations shall be generated within 3 seconds for a given query.
Scalability	The system shall support monitoring of at least 5,000 vehicles concurrently without degradation in dashboard responsiveness, and shall scale horizontally as fleet size grows.
Reliability	The platform shall maintain 99.5% uptime, with automatic failover for critical monitoring services.
Security	All data transmission between IoT devices and the cloud platform shall be encrypted (TLS 1.2+). Role-based access control shall restrict data visibility by user type.
Usability	The fleet manager dashboard shall be usable by non-technical staff with no more than 30 minutes of onboarding, following a clean, intuitive UI.
Maintainability	The system shall follow a modular microservice architecture allowing individual components (ML engine, routing engine, dashboard) to be updated independently without full-system redeployment.
Availability	Core monitoring and alerting services shall be available 24/7, including during scheduled maintenance of non-critical modules.
Interoperability	The system shall expose REST APIs to integrate with third-party charging network providers and traffic data services.
Portability	The web dashboard shall function correctly across major browsers (Chrome, Edge, Firefox, Safari) and the driver app shall support both Android and iOS.

Category	Requirement
Data Retention	Historical telemetry and charging data shall be retained for a minimum of 2 years for trend analysis and compliance.
Accuracy	The battery degradation prediction model shall maintain a minimum accuracy threshold of 85% against validated historical outcomes before being used for production alerts.

7. ACTORS AND USE CASES

7.1 Primary and Secondary Actors

Primary Actors:

- **Fleet Manager** – Monitors fleet health, views dashboards/reports, manages vehicle profiles, approves routing/charging strategies
- **Driver** – Receives route guidance, vehicle alerts, and charging instructions
- **Maintenance Technician** – Receives predictive maintenance alerts, logs completed service actions
- **System Administrator** – Manages user accounts, roles, and system configuration

Secondary Actors:

- **Charging Station System** – Supplies real-time charging slot availability
- **GPS/Traffic Data Provider** – Supplies live location and traffic condition data
- **ML Prediction Engine** – Runs battery degradation and maintenance prediction models

7.2 Use Case Descriptions

UC-1: Monitor Battery Health

- **Actors:** Fleet Manager; IoT sensors (trigger)
- **Precondition:** Vehicle IoT device is active and connected.
- **Main Flow:** (1) IoT sensor captures SoC, SoH, voltage, temperature. (2) Data is transmitted to the cloud backend. (3) System updates the vehicle's live status. (4) Fleet Manager views updated status on dashboard.
- **Alternate Flow:** If sensor connectivity is lost, system flags the vehicle as "offline" and notifies the Fleet Manager.
- **Postcondition:** Battery status is current and visible in real time.

UC-2: Predict Maintenance Need

- **Actors:** ML Prediction Engine; Maintenance Technician
- **Precondition:** Sufficient historical telemetry data exists for the vehicle.
- **Main Flow:** (1) ML engine analyzes recent and historical telemetry. (2) Model computes a degradation/failure probability score. (3) If the score exceeds a configured threshold, a maintenance ticket is generated. (4) Ticket is assigned to an available technician.
- **Postcondition:** A maintenance alert/ticket exists and is visible to the assigned technician.

UC-3: Optimize Charging Schedule

- **Actors:** Fleet Manager; Charging Station System
- **Precondition:** Charging station availability data is accessible.
- **Main Flow:** (1) System checks current battery SoC against required range for next trip. (2) System queries charging station availability and electricity tariff windows. (3) System computes the optimal charging time slot. (4) Recommendation is sent to the Fleet Manager for approval or auto-applied per configuration.
- **Alternate Flow:** If no charging slot is available within the required window, system escalates an alert to the Fleet Manager.
- **Postcondition:** A charging schedule is generated and communicated to the driver/vehicle.

UC-4: Recommend Energy-Efficient Route

- **Actors:** Driver; GPS/Traffic Data Provider
- **Precondition:** Vehicle GPS is active and a destination is provided.
- **Main Flow:** (1) Driver/system inputs destination. (2) System retrieves live traffic and road data. (3) System computes route options weighted by energy consumption, not just distance/time. (4) Optimized route is delivered to the driver's in-vehicle display.
- **Postcondition:** Driver receives an energy-optimized route.

UC-5: View Real-Time Dashboard

- **Actor:** Fleet Manager
- **Main Flow:** Fleet Manager logs in, system aggregates live vehicle locations, battery levels, and active alerts, and renders them on a consolidated dashboard view.

UC-6: Generate Analytical Reports

- **Actor:** Fleet Manager
- **Main Flow:** Fleet Manager selects a date range and report type; system compiles utilization, energy-efficiency, and cost data; report is rendered on-screen and made available for export (FR-15).

UC-7: Track Vehicle Location

- **Actors:** Driver; Fleet Manager
- **Main Flow:** GPS module transmits live coordinates; system updates the map view in real time for both driver and fleet manager interfaces.

UC-8: Manage Alerts and Notifications

- **Actors:** Fleet Manager, Driver, Maintenance Technician
- **Main Flow:** System monitors defined trigger conditions (low battery, predicted failure, station unavailability, geofence breach); when a condition is met, a notification is pushed to the relevant user role via app/SMS/email.

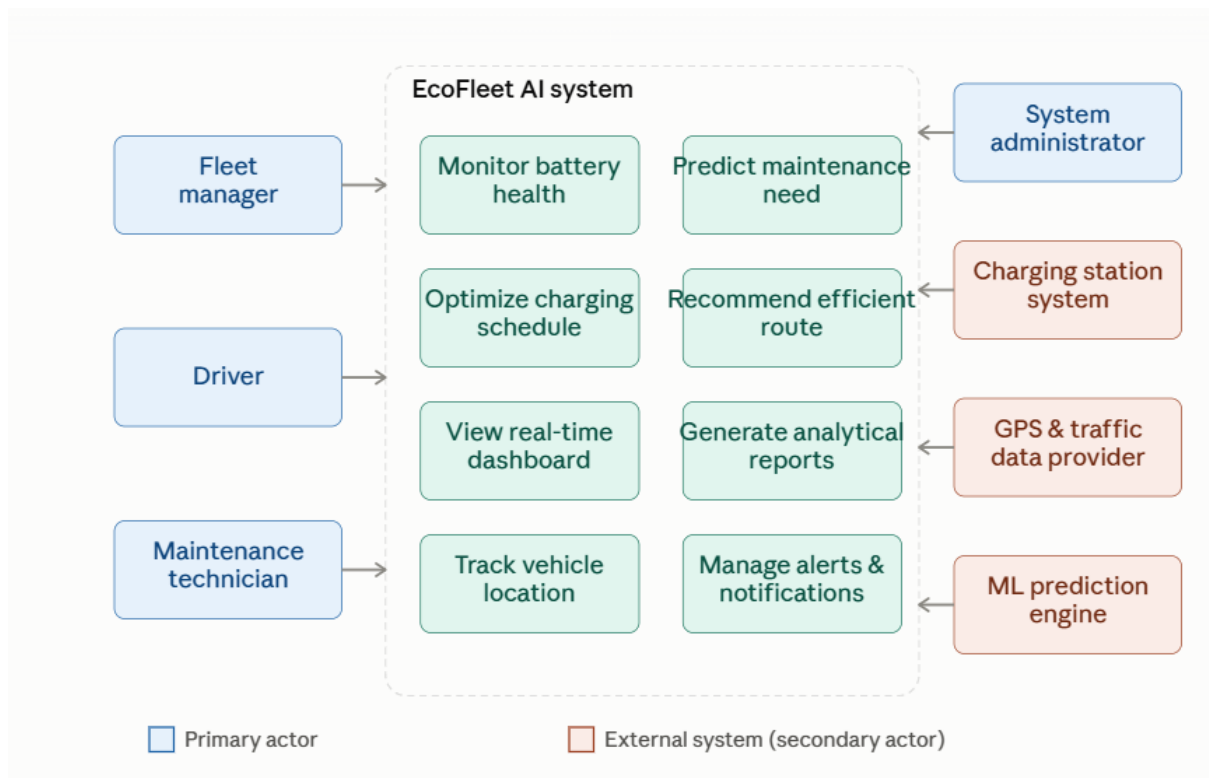
UC-9: Manage Vehicle Profile

- **Actor:** Fleet Manager
- **Main Flow:** Fleet Manager creates, edits, or retires vehicle records including registration, assigned driver, battery capacity, and service history.

UC-10: Configure System Settings

- **Actor:** System Administrator
- **Main Flow:** Administrator manages user roles, alert thresholds (FR-19), and integration credentials for external APIs.

8. Use Case Diagram



9. DATA REQUIREMENTS

Data Entity	Key Attributes
Vehicle	Vehicle ID, registration number, model, battery capacity, assigned driver, current status
Battery Telemetry	Timestamp, SoC (%), SoH (%), voltage, current, temperature, vehicle ID
Charging Session	Session ID, vehicle ID, station ID, start/end time, energy consumed (kWh), cost
Route	Route ID, origin, destination, waypoints, estimated energy consumption, traffic conditions used
Maintenance Record	Record ID, vehicle ID, predicted issue, severity, technician assigned, resolution status
User	User ID, name, role (Manager/Driver/Technician/Admin), credentials, contact info
Alert/Notification	Alert ID, type, severity, target user, timestamp, status (read/unread)
Charging Station	Station ID, location, slot availability, connector type, tariff rate

Data shall be stored in a structured cloud database with time-series support for telemetry, and shall support historical querying for trend analysis and ML model retraining. Telemetry data volume is expected to be high-frequency (per-vehicle updates every few seconds), so the database layer should support time-series-optimized storage (e.g., partitioned tables or a dedicated time-series database) separate from relational entities like Vehicle, User, and Maintenance Record.

10. EXTERNAL INTERFACE REQUIREMENTS

User Interfaces: Web-based fleet manager dashboard (desktop); mobile app for drivers and technicians with route guidance and alert notifications.

Hardware Interfaces: IoT battery management sensors (voltage, current, temperature) and GPS modules installed on each vehicle, communicating over cellular/LTE-M or Wi-Fi to the cloud backend.

Software Interfaces:

- REST API integration with third-party charging network providers for real-time slot availability
- REST API integration with a mapping/traffic data provider for route and traffic data
- ML model-serving API (internal microservice) for degradation/maintenance prediction
- Email/SMS/push notification gateway for alerts

Communication Interfaces: Encrypted HTTPS/TLS communication between IoT devices, mobile apps, and cloud backend; MQTT protocol recommended for lightweight, low-latency IoT telemetry transmission.

11. SYSTEM FEATURES

Real-Time Fleet Monitoring – Live view of battery health, location, and status for every vehicle in the fleet.

- **Predictive Maintenance Engine** – ML-driven early warning system for battery degradation and component failure.
- **Smart Charging Optimizer** – Automated scheduling that balances battery needs against electricity cost and station availability.
- **Energy-Efficient Route Planner** – Route recommendations that minimize energy usage, not just travel time.
- **Fleet Analytics & Reporting** – Historical and real-time reports on utilization, efficiency, and cost.
- **Role-Based Access & Alerts** – Tailored dashboards and notifications for managers, drivers, technicians, and administrators.

- **Geofencing & Route Deviation Alerts** – Automatic notification when a vehicle leaves its assigned operating zone.
- **Audit and Compliance Logging** – Full historical record of charging sessions and maintenance actions for reporting.

12. ASSUMPTIONS AND CONSTRAINTS

Assumptions:

- All fleet vehicles are equipped with, or can be retrofitted with, compatible IoT battery and GPS sensors.
- Charging station operators expose an API or data feed for real-time availability.
- Fleet operators have reliable internet/cellular connectivity in their operating regions.
- Historical vehicle data will be available in sufficient volume to train reliable ML prediction models.

Constraints:

- **Technical:** The system depends on third-party GPS/traffic and charging-network APIs; unavailability or rate-limiting of these APIs directly affects route and charging recommendations.
- **Operational:** Fleet managers must approve or override AI-generated charging/route recommendations in early deployment phases to build trust in the system.
- **Economic:** Development and IoT hardware retrofitting costs may limit initial deployment to a pilot subset of the fleet.
- **Legal/Compliance:** The system must comply with regional data privacy regulations regarding driver location tracking.
- **Time:** The predictive maintenance model requires a minimum historical data collection period (estimated 2–3 months) before reliable predictions can be generated.

13. RISK ANALYSIS

Risk	Impact	Mitigation
Third-party API downtime (charging/traffic data)	Route/charging recommendations become unavailable or stale	Implement caching and fallback to last-known-good data
Insufficient historical data for ML model	Inaccurate maintenance predictions, false alerts	Delay model activation until minimum data threshold is met; use rule-based fallback initially
IoT sensor connectivity loss	Gaps in real-time monitoring	Flag vehicle as offline; alert fleet manager; buffer data on-device for later sync
Data privacy non-compliance	Legal exposure, loss of stakeholder trust	Encrypt data in transit/at rest; anonymize location data where feasible

14. REQUIREMENT VALIDATION(Completeness & Consistency)

Completeness Check: All eight platform objectives from the problem statement (monitoring, prediction, charging optimization, route recommendation, dashboards, reporting, productivity, cost/emission reduction) are traceable to specific functional requirements (FR-1 through FR-20) and use cases (UC-1 through UC-10), confirming stakeholder requirements have been addressed.

Consistency Check: No conflicting requirements were identified. Charging optimization (FR-5) and route optimization (FR-6) are designed as independent but data-sharing modules, avoiding overlap. Alert generation (FR-9) is consistently triggered from the prediction engine (FR-4), charging module (FR-5), and geofencing (FR-18), ensuring uniform notification behavior across the system.

Ambiguity/Redundancy Review: Terms such as "real-time" and "energy-efficient" were quantified with specific thresholds (5-second update latency, minimized kWh/km) in the Non-Functional Requirements section to remove ambiguity. No redundant requirements were found across the functional list.

Suggested Improvements:

- Add a feedback loop requirement where actual post-trip energy consumption is compared against predicted consumption to continuously improve the ML route model.
- Define a formal SLA requirement for third-party charging station API response times, since charging optimization accuracy directly depends on this external data.
- Add an explicit data-retention and anonymization requirement to strengthen compliance readiness before final design sign-off.

15. GLOSSARY

Battery degradation – The gradual, irreversible reduction in a battery's maximum charge capacity over time and use.

- **Geofence** – A virtual geographic boundary that triggers a system alert when crossed.
- **Predictive maintenance** – Maintenance scheduled based on forecasted failure rather than fixed time intervals.
- **Tariff window** – A time period during which electricity is priced differently (peak/off-peak).

16. CONCLUSION

EcoFleet AI translates the fragmented, tracking-only approach of conventional fleet software into a unified, predictive, and optimization-driven platform. By combining IoT telemetry, machine learning-based predictive maintenance, smart charging scheduling, and energy-aware routing, the system directly addresses the operational pain points described in the problem statement — unplanned downtime, inefficient charging, and suboptimal routing — while supporting the broader goal of sustainable, cost-effective electric fleet operations. Beyond solving these immediate operational challenges, the platform establishes a foundation that fleet operators can build on as their EV adoption scales, since every additional vehicle, sensor, and charging station brought online strengthens the predictive models and improves recommendations across the entire fleet rather than adding operational risk.

Ultimately, EcoFleet AI is designed not as a one-time software deployment but as a long-term operational partner for fleet managers — one that grows alongside the organization's transition to electric mobility, adapts to new regulatory and sustainability requirements, and continuously narrows the gap between what fleet operators currently know about their vehicles and what they actually need to know to run an efficient, low-emission, and resilient fleet.

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